MICROBIOLOGY

CHAPTER 26 :

FUNGI

Done by:
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Reference:

Slide and

Dr sereen notes

Chapter 26: Fungi - The last chapter (♡˙︶˙♡) ♡＼(￣▽￣)／♡

<u>Fungi or Eumycota</u> : Eu→ true
Mycota→ fungi

- <u>Eukaryotic</u>, spore-bearing
- Chemoorganoheterotrophs with absorptive metabolism

Energy source:	Electron source:	Carbon source:
Oxidation of organic or inorganic compounds	Organic molecules	Reduced, preformed, organic molecules from other organisms

- <u>Saprophytes</u>

➔ Meaning that they absorb nutrients from dead organic material by releasing degradative enzymes

  - osmotrophy - absorb soluble products

- Lack chlorophyll→so they don't do photosynthesis
- Reproduce sexually and asexually.

وهاد جدا ضروري بعملية تصنيفهم والمجموعات الي حندرسها هلأ لازم تكونوا عارفين كل وحدة بتتكاثر بشكل جنسي او لا جنسي

<u>Terminology</u>:

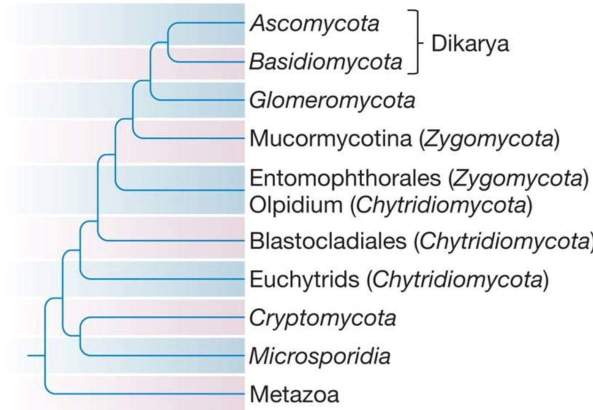
- اول واهم نقطة انو لازم تعرف وين ما تلاقي هاد المقطع -Myco اعرف انو عم نحكي عن الفطريات(fungi)

- <u>Mycology</u> : study of fungi
- <u>Mycologists</u> : scientists who study fungi
- <u>Mycoses</u> : diseases caused by fungi
- <u>Mycotoxicology</u> : study of fungal toxins and their effects

Taxonomy of Fungi:

- 90,000 fungal species have been described, possible 1.5 million
- Six major fungal groups:
    1. Chytridiomycota
    2. Zygomycota
    3. Glomeromycota
    4. Ascomycota
    5. Basidiomycota
    6. Microsporidia

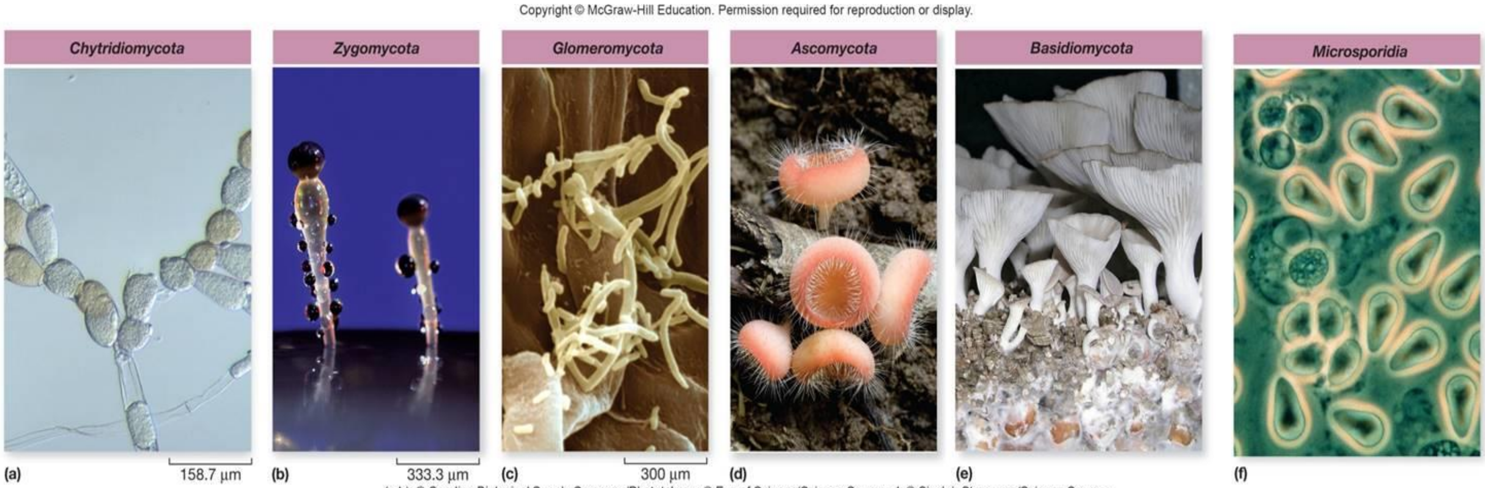
➔ They are classified according to the type of spores they produce and their sexual lifecycles



Source: "Reconstructing the early evolution of Fungi using a six-gene phylogeny" by James, T.Y. et al., *NATURE*, Vol. 443, 2006, p. 818–822

Tricky Taxonomy of Fungi(مو مهم كتير)

- Basidiomycota and Ascomycota are dikarya
    - two parental nuclei are initially paired
    - nuclei fuse, undergo meiosis, produce haploid progeny
- Zygomycota and Chytridiomycota are paraphyletic
    - taxonomic group includes some descendents of a single common ancestor



(a,b): © Carolina Biological Supply Company/Phototake; c: © Eye of Science/Science Source; d: © Sinclair Stammers/Science Source; e: © Robert W. Seagull; f: Dr. James Becnel, USDA ARS Gainesville, FL and Society of Invertebrate Pathology

Fungal Distribution and Importance

- Primarily terrestrial, few aquatic(يعني بشكل اساسي توجد بالارض مو بالهوا او بالمي بس عدد قليل منلاقيه بالمي)
  - They have a global distribution from polar to tropical regions (منتشرين بكل ارجاء العالم حتى بالاقطاب)
- Many are pathogenic in plants or animals
  → One very important thing about fungi is that they secret very powerful toxins
  - They can cause diseases for us like superficial infections(الامراض الجلدية) or systematic diseases→in the blood
    - They cause a disease called athlete's foot
    - Some cause diseases under the nails
    - Razors(شفرات الحلاقة) can also be infected with fungi and they cause diseases if not cleaned probably
- Some form associations
  - mycorrhizae – associations with plant roots → Remember: Myco→relates to fungi
    Rihzo→relates to roots
  - lichens (chapter 20)
    - associations with algae or cyanobacteria
      → Lichens can't live without each other : fungi can't live without algae\ cyanobacteria and algae\ cyanobacteria can't live without fungi
- Decomposers
  - degrade complex organic material in the environment to simple organic compounds and inorganic molecules

الهم دور كبير بتحلل الجثث بالتربة عن طريق تحويل المواد العضوية الي فيها لمواد عضوية ابسط ومواد غير عضوية وتزكروا انو مو بس الfungi بتلعب هاد الدور، اخدنا بكتيريا برضو بتعمل نفس الوظيفة متل: actinomycetes and arthrobacter

  - carbon, nitrogen, phosphorus, and other critical constituents are recycled for other living organisms

وطبعا كل هاي المواد الي تم انتاجها رح ترجع لكائنات تانية ليستهلكوها (توازن جميل (o˘◡˘o))

  - Have an important role in biodeterioration*

* The process of biodegradation can be divided into three stages: biodeterioration, biofragmentation, and assimilation
Biodeterioration is sometimes described as a surface-level degradation that modifies the mechanical, physical and chemical properties of the material.
Biodeterioration is defined as the undesirable action of living organisms on Man's materials, involving such things as breakdown of stone facades of buildings,[6] corrosion of metals by microorganisms or merely the esthetic changes induced on man-made structures by the growth of living organisms.
هاي المعلومة خارجية بس الدكتورة حكت ابحثوا عنها..ف ما تحفظوا اشي4ولكن بكفي انكم تفهموا الفرق بشكل العام. الbiodeterioration هو اول مرحلة من الbiodegradation وفي باحث سماه العمل الغير مرغوب فيه لصناعات الانسان لانو هاي العملية بتخرب اشياء كتير

Fungal Distribution and Importance – Industrial and Research Use

- Industrial importance
  - fermentation – yeast used in making bread, wine, beer, cheese, soy sauce
  - organic acids – citric and gallic acid
  - certain drugs – ergometrine, cortisone
  - antibiotics – penicillin, griseofulvin
  - immunosuppressive agents (suppress the immune system in transplantation surgeries) – cyclosporin

- Research use
  - geneticists, cytologists, biochemists, biophysicists, and microbiologists use it
  - Saccharomyces cerevisiae: The most used kind of fungi in research (هي نفسها الخميرة تبعت المخبوزات)
    - yeast model system for cell biology, genetics, and cancer

Fungal Structure

- their cell walls are composed of chitin polysaccharide
  (and remember *Streptomycetes* have chitinase which can be transformed to plants to be resistant to fungi by breaking their cell walls)
- Single-celled microscopic fungi = yeasts
- The body or vegetative structure of a fungus is called a **thallus**
  - multicellular masses are called molds.
    (يعني الfungi الي بخلية وحدة منحكيلو yeast والي بكون مكون من اكتر من خلية منسميهmold)
  - The thallus of a **mold** consists of long, branched, threadlike filaments called **hyphae** that form a tangled mass called a **mycelium**.
  - للتذكير..هاي النقطة انشرحت من قبل. التركيب الي بشبه الخيط منسميه hypha <-- مجموعة من الhypha بتعطي hyphae<-- مجموعة الhyphae بتعطي mycelium <-- مجموعة الmycelium بتعطي mycelia<-- ومجموعة الmycelia بتعطيthullus

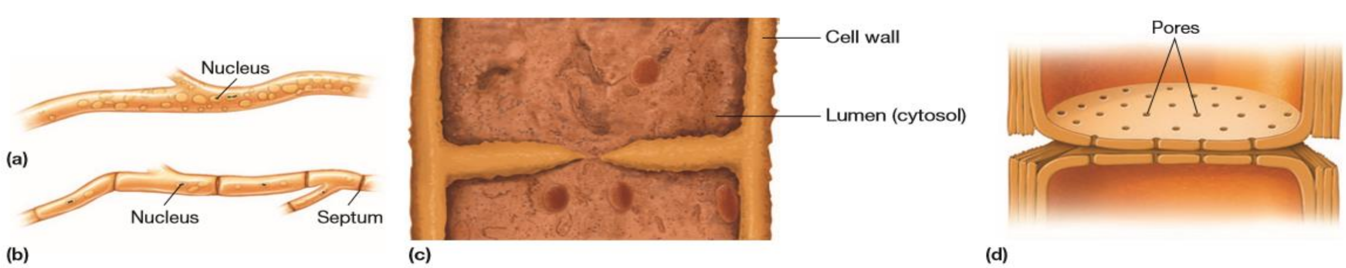


Figure 26.3 Hyphae. Drawings of (a) coenocytic hyphae (aseptate) and (b) hyphae divided into cells by septa. (c) A section of a representative fungal filament demonstrates that in some species, cellular compartments communicate by a single pore. (d) Drawing of a multiperforate septal structure.

We have two types of hyphae:

1- non septated and multinucleated hyphae and does not have septa (no boundaries) (s., septum) These hyphae are called **coenocytic or aseptate hyphae** (figure 26.3a).
2- septated hyphae→ have septa (boundaries) that separates between cells and each cell has one nucleus
   these might have one pore in their septa like in (figure 26.3c) or multiple pores like in (figure26.3d)

Fungal Reproduction

A. Asexual: Parent cell undergoes mitosis to form daughter cells
   - Mitosis in vegetative cells may be concurrent with budding to produce a daughter cell
   - May proceed through a spore form

   Types of asexual reproduction:
   a) Arthrocnidia (arthrospores)
   b) Sporangiospores
   c) Coindiospores
   d) Blastospores

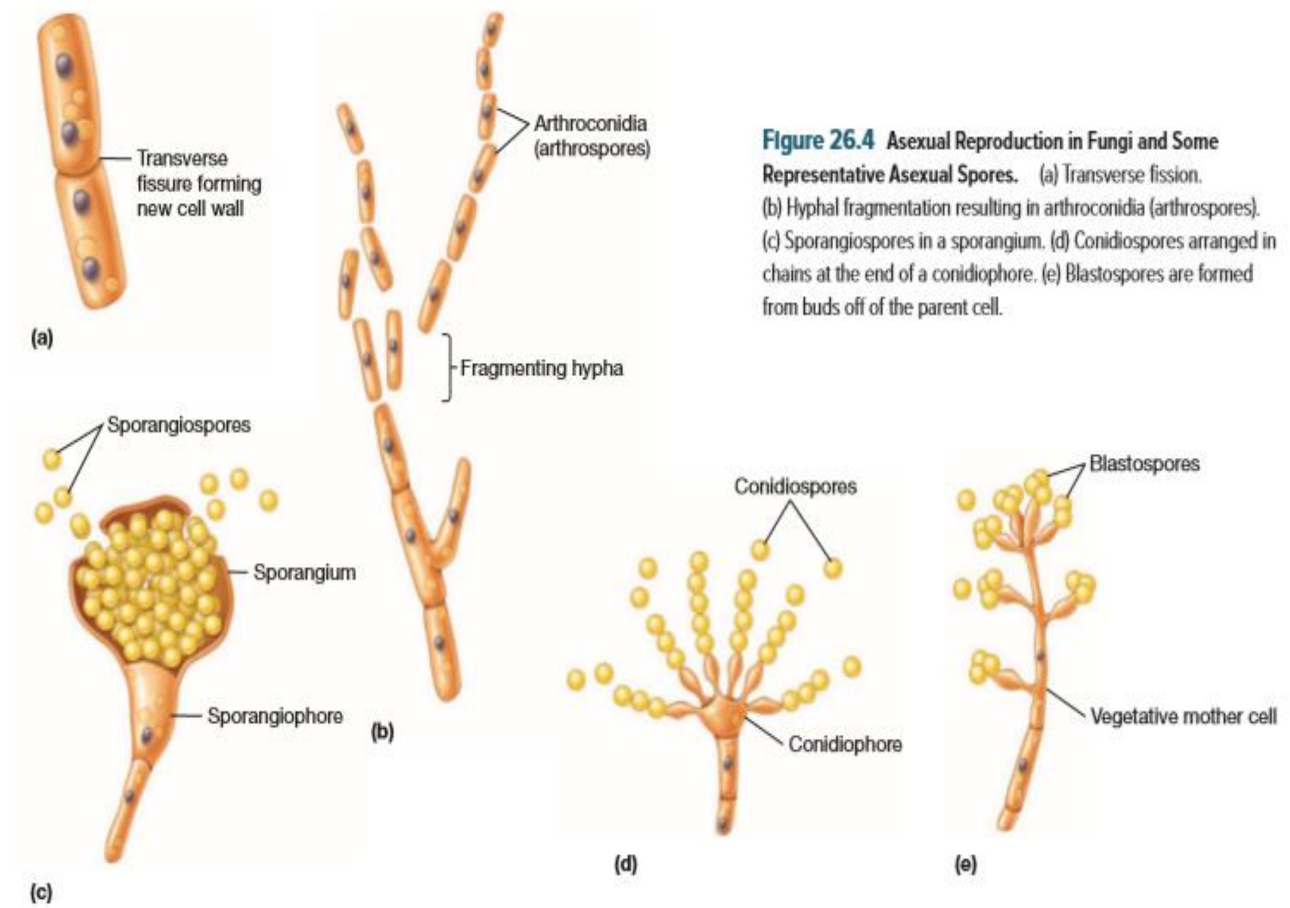


Figure 26.4 Asexual Reproduction in Fungi and Some Representative Asexual Spores. (a) Transverse fission. (b) Hyphal fragmentation resulting in arthroconidia (arthrospores). (c) Sporangiospores in a sporangium. (d) Conidiospores arranged in chains at the end of a conidiophore. (e) Blastospores are formed from buds off of the parent cell.

B. Sexual reproduction: Involves <u>fusion</u> of compatible nuclei
  - Homothallic: Sexually-compatible gametes are formed on the same mycelium (self-fertilizing)
  - Heterothallic: Require outcrossing between different, yet compatible mycelia
    →a dikaryotic stage can exist temporarily prior to fusion of two haploid nuclei

هاي الفقرة بس مروا عليها سريع..الدكتورة ما بدها تفاصيل

<u>Fungal Groups</u>

1. *Chytridiomycota*: Produce Motile Spores

- Simplest fungi, also called chytrids
  - free living, saprophytic (live on dead decayed organic matter)
  - parasitic forms infect aquatic plants and animals, insects
  - found in the anoxic rumen of herbivores
  - may be responsible for large-scale mortality of amphibians including over 100 species of frogs native to tropical America
- Produce a zoospore with single, posterior, whiplash flagellum
  - most primitive form of spore dispersal
  - flagella lost in higher forms
- <u>*Asexual and sexual reproduction*</u>
- Many members degrade cellulose and keratin

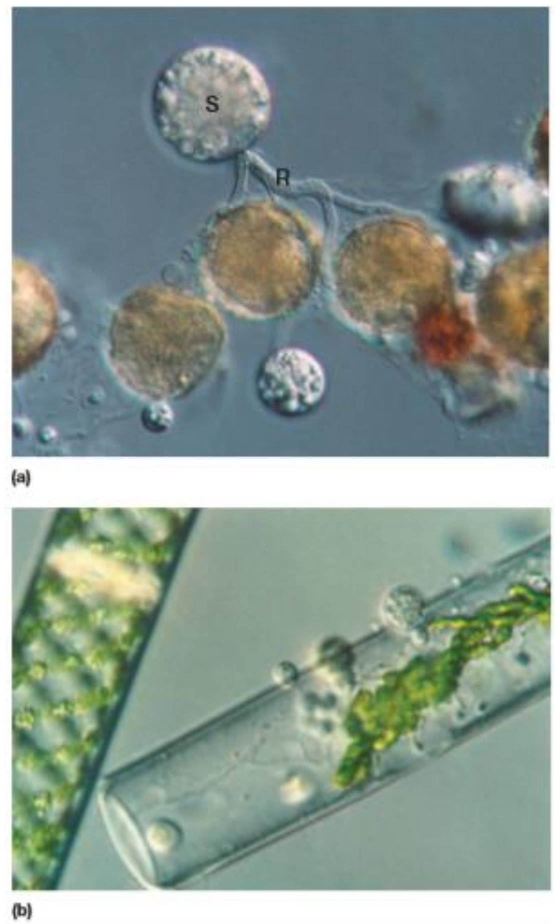


Figure 26.6 *Chytridiomycota.* (a) A chytrid with a single round sporangium (S) that is about 50 μm in diameter. In this image, the fungus is growing between brown pollen grains, which it will eventually penetrate with its branched rhizoids (R). (b) Parasitic chytrids attached to the surface of a photosynthetic protist (a green alga).

MICRO INQUIRY *What is the function of the chytrid rhizoids?*

2. *Zygomycota*: Fungi with Coenocytic Hyphae

- Most are saprophytes
  - a few are plant and animal parasites
- Form coenocytic hyphae(Non-septate hyphae) containing numerous haploid nuclei
- Some of industrial importance
  - foods, antibiotics and other drugs, meat tenderizer, and food coloring
- Usually reproduce asexually by spores that develop at the tips of aerial hyphae
- Sexual reproduction occurs when environmental conditions are not favorable
  - requires compatible opposite mating types

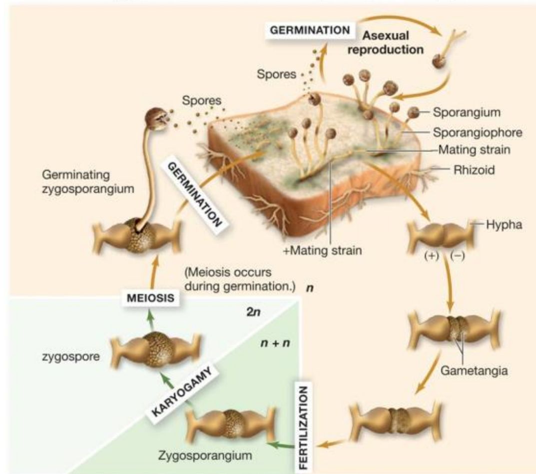
- hormone production causes hyphae to produce gametes
- gametes fuse, forming a zygote
- zygote becomes zygospore

*Rhizopus: don’t mix it up with rhizoBIUM :the bacteria that do nitrogen fixation at the roots of legumes

➔ *Genus Rhizopus**

R. stolonifer: is a common member of this group.

- grows on surface of moist carbohydrate rich foods such as bread
- hyphae quickly cover surface as rhizoids, absorb nutrients
- stolon hyphae become form new rhizoids



ما في داعي تدرسوا هاي الصورة مو مطلوبة بس المهم انكم تتزكروا انو هاد النوع هو عفن الخبز

- Importance of Genus Rhizopus:
  - Rhizopus-Burkholderia symbiosis
    →seedling blight in rice bacterium
    Burkholderia growing within Rhizopus produces toxin

الBurkholderia هي نوع بكتيريا والRhizopus هو الفطر الي عم نحكي عنو. في مرض (seedling blight) كانوا مفكرين انو هاد الفطر هو الي بسببو ولكن مع الدراسات وجدوا انو في علاقة تعايش بينو وبين بكتيريا اسمها Burkholderia عايشة جواه هي الي بتسبب هاد المرض عن طريق افراز بعض الtoxins.

  - Used to produce tempeh from soybeans
  - Used with soybeans to make sufu curd
  - Commercially:
    used to produce anesthetics, birth control, alcohols, meat tenderizers, yellow coloring in margarine

3. *Glomeromycota:* Are Mycorrhizal Symbionts

- Aseptate flat hyphae (appressoria) to penetrate host plants; produce large, multinucleate spores and only reproduce asexually
- Major importance as mycorrhizal symbionts of vascular plants(النباتات الوعائية)
  - form intracellular associations within roots of almost all herbaceous plants and tropical trees and they might be in two forms:
    - Ectomycorrhizae →do not penetrate root cells
    - Arbuscular mycorrhizae→ penetrate root cell wall

- mutualistic relationship (both benefit)
  - fungus helps protect host from stress, delivers soil nutrients to the plant; plant provides carbohydrates to fungus

4. *Ascomycota* :Includes Yeasts and Molds

(a) *Morchella esculenta* (b) *Tuber brumale*
a: © Juniors Bildarchiv GmbH/Alamy; b: © Kelly Cline/E+/Getty Images

- sac fungi: because they have a reproductive sac-like structure called ascus(a sac) and this ascus holds endospores
- found in freshwater, marine, and terrestrial habitats
- red, brown, and blue-green molds cause food spoilage (The pink bread mold *Neurospora crassa* عفن الخبز ,is an important research tool in genetics and biochemistry عفن الخبز)
- some are human and plant pathogens
- some yeasts and truffles are edible
- some used as research tools
- Ascomycota Yeast Life Cycle:

ONLY ONE THING REQUIRED: they do mitosis and budding in nutrient rich environments and meiosis in nutrient poor environments

- Alternates between haploid and diploid
  - in nutrient rich, mitosis and budding occurs at non-scarred regions
    - stops after entire mother cell is scarred
  - nutrient poor, meiosis and haploid ascus containing ascospores formed
    - haploid cells of opposite mating types fuse
    - tightly regulated by pheromones
- Filamentous Form Life Cycle:
  - Asexual reproduction → associated with the production of externally produced asexual spores called **conidia**
  - Sexual reproduction
    - ascus formation with ascospores
    - opposite mating types form zygote
    - ascospores forcefully released from ascocarp, germinate

Figure 26.10 Asexual Reproduction in Filamentous Ascomycetes. Characteristic conidiospores of *Aspergillus* spp. as viewed with the scanning electron microscope.

- what makes Ascomycota special is that it has a structure called **sclerotia**→compact masses of hyphae survive the winter and stressful conditions then germinate

➔ *Aspergillus*: a genus of Ascomycota

- *A. fumigatus*
  - ubiquitous environmental , commonly found in homes and the workplace
  - triggers allergies and is implicated in the increased incidence of severe asthma and sinusitis. It is also pathogenic, infecting immunocompromised individuals with a mortality rate of nearly 50%
- *A. oryzae*
  - production of traditional fermented foods and beverages in Japan , including saki and soy sauce.
  - important in biotechnology
- *Aspergillus*
  - 37 Mb genome, model system

- More Pathogenic Ascomycota:
  - *Claviceps purpura*
    - parasite on higher <u>plants</u>
    - **Ergotism**
      - the toxic, delusional condition in humans and animals that eat grain infected with the fungus, is often accompanied by gangrene, nervous spasms, abortion, and convulsions.
      - The pharmacological activities are due to an active ingredient, lysergic acid diethylamide (LSD).
      - over 40,000 deaths from ergot poisoning were recorded in France in the year 943

Figure 26.12 **Ergot of Rye.** The ascomycete *Claviceps purpurea* infects rye and other grasses, producing hard masses of hyphae known as ergots in place of some of the grains (fruits). Ergots produce alkaloids that cause psychotic delusions in humans and animals when products made with infected grain are consumed. Ergots have been used to treat migraine headaches and hasten childbirth.

  - *Candida, Blastomyces, Histoplasma*
    - human pathogens
  - *Stachybotrys* – "sick building syndrome"
  - *Aspergillus* – aflatoxins (منلاقيهم بالتفاح الامريكي والمكسرات الي الهم مدة)
    and cancer

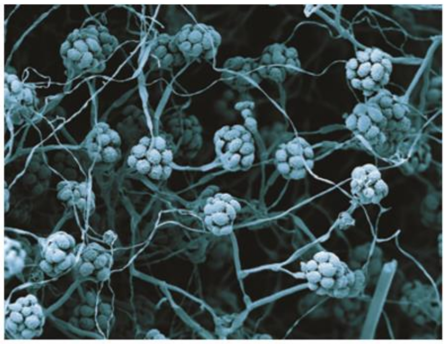


Figure 26.13 **Fungal "Sick Building Syndrome" Can Be Caused by an Ascomycete.** Scanning electron micrograph of dense mycelia and conidiophores characteristic of *Stachybotrys chartarum*.

5. *Basidiomycota* :Includes Mushrooms and Plant Pathogens

- Basidiomycetes (club fungi)
  - examples include rusts, shelf fungi, puffballs, toadstools, mushrooms
  - <u>sexual reproduction</u> form basidium
    - basidiospores are released at maturity
- Decomposers:
  →most of them are saprophytes they decompose plant debris especially cellulose and lignin and because of that they are called decomposers

- Edible→ *Agaricus campestris*
- non-edible mushrooms
  - Toxins are poisons or hallucinogens

- Pathogens of humans, other animals, and plants
  - e.g. Cryptococcus neoformans – cryptococcosis
    - systemic infection, primarily of lungs and central nervous system

Figure 26.15 ***Ustilago maydis.*** This pathogen causes tumor formation in corn. Note the enlarged tumors releasing dark teliospores in place of normal corn kernels.

- Urediniomycetes and Ustilaginomycetes
  - Basidiomycetes known as urediniomycetes and ustilaginomycetes include important plant pathogens causing "rusts" and "smuts," respectively.
    - dimorphic (plant-associated fungi grow in the mycelial form but are yeast like in the external environment)
      - yeasts are infectious form
      - mold - teliospores form tumors
  - Some urediniomycetes are also human pathogens

- Zygomycota, Ascomycota, basidiomycota →from their names we will know the name of sexual spores they produce
  - Zygomycota→ zygo spores
  - Ascomycota→Asco spores
  - Basidiomycota→basidiospores

6. *Microsporidia* :Intracellular Parasites

Of all the fungi, members of Microsporidia have the most confused taxonomic history. These microbes were considered protists. However, analysis of ribosomal RNA and specific proteins such as α- and ß-tubulin shows that they are fungi.

- Obligate intracellular fungal parasites that infect insects, fish, and humans
  - Aquatic birds are common hosts and contribute to large numbers of spores in environment
- Transitional form is a spore structure capable of surviving outside the host
- Structurally similar to 'classic' fungi
  - contain chitin, trehalose, and mitosomes
  - however, lack mitochondria, peroxisomes and centrioles
  - unique morphology is polar tube essential for host invasion
- Pathogensis:
  - Human infections
    - *Enterocystozoon bieneusi*
      - diarrhea
      - Pneumonia
    - *Encephalitozoon cuniculi*
      - encephalitis →in the brain
      - Nephritis→in kidneys
    - They are particularly problematic for immunocompromised individuals, especially those with HIV/AIDS.

Additional notes:

→Fungi are incredibly important to life on this planet, as they recycle nutrients from decomposition processes but, they can sometimes grow in/on places we don't want them to, causing diseases of plants and animals

→The only group of fungi that reproduce just asexually is called *Deuteromycota*

The End

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